



Break It Into Parts

Big things are made of small parts

Name:

Date:

★ I can break a big picture into the small shapes that make it.

Big things are made of parts

Look closely and you will see that big things are built from smaller shapes. Breaking a big thing into its parts is called DECOMPOSITION.

Watch Bit — a rocket is made of 2 parts



top



body

Bit's rocket is a triangle on top of a square. That is 2 parts.

1 A house — draw its 2 parts

A house is made of a square ■ (walls) and a triangle ▲ (roof). Draw the two parts, one in each box.

2 A snowman — draw its 3 parts

A snowman is made of three circles ● ● ●. Draw the three parts, one in each box.

3 You try — Bit the robot

Bit is made of 4 parts: a square ■ body, a circle ● head, and two feet (each a square ■). Draw all four parts, one in each box.

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Big idea

Breaking a big thing into small parts is called decomposition. Coders do it too — they break a big problem into small, easy steps.



Break It Into Parts

Quick facilitation notes & answer key

What this worksheet teaches

Children see that a whole picture is built from smaller shape parts, and draw those parts. Breaking a whole into parts is a first thinking-skill behind coding — splitting a big job into smaller pieces. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil or crayon. You read each prompt aloud; the child draws the parts. Optional: pattern blocks to build a house or rocket and then take it apart into its shapes.

How to lead it (about 15 minutes)

1. Show them (you do). Show the rocket — "a triangle on top, a square for the body. That is 2 parts."
2. Do one together. Exercise 1 — the house is a square (walls) and a triangle (roof).
3. Let them try. Children draw the snowman's 3 circles (Ex 2), then Bit the robot's 4 parts (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — house = 2 parts (1 square + 1 triangle). Exercise 2 — snowman = 3 parts (3 circles).

Exercise 3 — Bit the robot = 4 parts (3 squares + 1 circle).

If children get stuck — and ways to adjust

Watch for: drawing the whole picture instead of separate parts. Point to each empty box as one part.

Make it easier: build the object with blocks, then lift each shape out and name it.

Make it harder: ask what other things are made of parts (a face, a flower, a car).

Make it active: have children use their bodies to make a shape, then combine into a group picture.

Ontario Kindergarten link (official wording)

Problem Solving and Innovating — children analyze how a whole is composed of parts.

In plain terms: seeing that a whole picture is built from smaller parts.

No reading is needed from the child — you read each prompt aloud; the child answers by drawing or circling.

You'll know it worked when

The child draws the correct number of separate shape parts for each whole (2, then 3, then 4).